How Pruning Criteria Shape Post-RLVR Recovery: A Controlled Study of Structured MLP Pruning in a Reasoning LLM

Anonymous ACL submission

Abstract

Reinforcement learning with verifiable rewards (RLVR) is widely regarded as essential for recovering pruned reasoning models, substantially outperforming supervised fine-tuning (SFT). We find that this advantage vanishes under fine-grained structured pruning: across four criteria applied to a 7B reasoning model at moderate sparsity, SFT and RL recover to the same accuracy regardless of criterion. We term this *reward-accuracy decoupling*. What the pruning criterion does control is RL training dynamics. To study this, we introduce Recovery-Aware Structured Pruning (RASP), a scoring function that balances gradient importance with representational diversity via Centered Kernel Alignment (CKA). The resulting *recovery capacity spectrum* shows that diversity-preserving criteria sustain RL optimization while importance-dominated criteria fail entirely, yet none of these differences translate into accuracy gains over SFT. These findings reframe the role of RL in pruning recovery: SFT warmup captures most recoverable reasoning capacity, while RL serves as a diagnostic that reveals structural deficiencies invisible to accuracy. Directional trends hold across three benchmarks, with preliminary cross-scale and cross-model evidence[1].

1 Introduction

Structured pruning compresses reasoning Large Language Models (LLMs) by removing entire channel groups, but the pruned model must recover through fine-tuning. Ji et al. (2025) show that reinforcement learning with verifiable rewards (RLVR) substantially outperforms supervised fine-tuning (SFT) for this recovery. This positions RLVR as the dominant paradigm and creates a testable prediction: a pruning criterion that preserves the model's capacity for RL exploration should yield higher final accuracy. The evidence, however, comes from unstructured magnitude pruning at 96.5% parameter retention; whether the RLVR advantage holds under structural compression, which removes entire computational units and alters the architecture, remains untested.

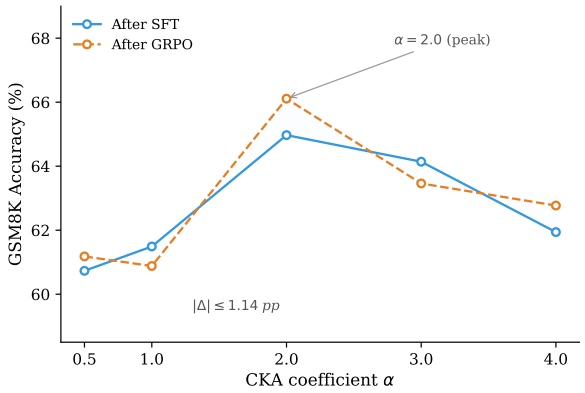


Figure 1: Dose-response curve for the CKA diversity coefficient α at 20% MLP channel sparsity on GSM8K (N=1,319). Recovery accuracy follows an inverted-U, peaking at α=2.0 (66.11%). SFT and GRPO curves nearly overlap ($|\Delta| \leq 1.14$ pp at every point), illustrating reward-accuracy decoupling.

Our experiments reveal that it does not. On DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025) pruned to 20% MLP channel sparsity, one criterion achieves 4.4 times the peak RL reward of another (CKA-only vs. Standard) under GRPO (Shao et al., 2024), yet for each criterion individually, SFT and GRPO recover the model to within 1 percentage point of each other. The pruning criterion determines whether RL training succeeds or collapses, but this stark difference leaves almost no trace in final accuracy. We term this pattern *reward-accuracy decoupling*.

This disconnect is surprising in light of recent mechanistic work. RL compresses incorrect reasoning paths while SFT expands correct ones (Mat-

[1]Code and data are available at https://anonymous.4open.science/r/recovery_aware_struct_prune-5FA5/.

sutani et al., 2026). These asymmetric pressures predict that the two paradigms should complement each other, not converge to the same outcome. Yet existing structured pruning methods for reasoning LLMs (Wang et al., 2026, 2025; Ma et al., 2023; Kolawole et al., 2024) optimize for immediate post-pruning quality without examining how the criterion shapes RL recovery. No prior work has compared multiple criteria under RL or tested whether the "RLVR > SFT" advantage persists across pruning strategies.

To investigate systematically, we introduce Recovery-Aware Structured Pruning (RASP), a scoring function that combines gradient importance with CKA-based representational diversity (Kornblith et al., 2019). A single coefficient α controls the balance (Figure 1): the dose-response curve reveals an inverted-U in recovery accuracy, peaking at α=2.0 (66.11% on GSM8K; Cobbe et al., 2021). Under GRPO, the four criteria span a **recovery capacity spectrum** that tracks preserved diversity rather than importance: criteria that maximize diversity sustain RL training, while importance-dominated criteria fail to learn. The SFT and GRPO curves in Figure 1 nearly overlap at every α, which confirms decoupling across the full spectrum. A depth pruning baseline (ShortGPT; Men et al., 2024) is the sole exception (-1.36 pp on GSM8K), which suggests that pruning granularity interacts with RL trainability.

Our contributions are:

1. **Reward-accuracy decoupling.** At 20–25% fine-grained MLP sparsity, SFT and GRPO yield equivalent accuracy regardless of pruning criterion, confirmed by TOST equivalence testing for CKA-only and Standard, with all criteria within comparable margins. At 15%, GRPO amplifies the CKA-guided advantage, which delineates the boundary (§4.3).
2. **Recovery capacity spectrum.** RASP reveals that four pruning criteria produce qualitatively distinct RL dynamics. Dose-response analysis traces an inverted-U in recovery, with preliminary evidence across scales and models within the Qwen2.5 family (§3.3, §4.2, §4.5).
3. **SFT warmup as the dominant recovery mechanism.** Across all criteria and sparsity levels, SFT warmup recovers from near-zero post-prune accuracy to 51–64%, while subsequent GRPO adds less than 1 pp. This dominance persists at 300 training steps. RL reward trajectories instead serve as structural diagnostics that reveal pruning deficiencies invisible to accuracy (§4.3, §4.5).

2 Related Work

Structured pruning of LLMs. Width pruning methods (LLM-Pruner (Ma et al., 2023), Bonsai (Kolawole et al., 2024), GRASPrune (Wang et al., 2026)) remove attention heads or MLP channels while preserving dense computation. Depth pruning (Men et al., 2024; Ashkboos et al., 2024; Yang et al., 2024b) removes entire layers but causes severe reasoning degradation (Shrestha et al., 2026); Monjur et al. (2026) revisit the trade-off under test-time scaling. For reasoning models, RESP (Wang et al., 2025), RAC (Lucas et al., 2025), and SSGR (Xiang et al., 2025) refine calibration data but retain gradient-based importance as the scoring criterion; Ding et al. (2026) confirm width pruning is more robust than depth pruning for reasoning but do not study RLVR recovery. A separate line uses functional structure to guide pruning: FANG (Yu et al., 2025) clusters FFN neurons by activation, FuncNet (Liu et al., 2025a) decomposes LLMs via independent component analysis, and MPruner (Hu et al., 2024) applies inter-layer CKA (Kornblith et al., 2019) for depth pruning in CNNs. We differ in both granularity and objective: *intra-layer* CKA between individual structures, optimizing for RL recovery capacity rather than immediate accuracy.

Pruning-aware recovery. Ji et al. (2025) show that RLVR restores pruned reasoning models to near-original accuracy, outperforming SFT, but evaluate only importance-based unstructured pruning at low sparsity (96.5% retention). Under fine-grained structural pruning at 20–25% sparsity, our controlled experiments indicate that this advantage diminishes when pruning preserves representational diversity, the effect may thus reflect differences in experimental setup rather than an inherent property of RL. Other work optimizes complementary axes: post-pruning scaling (P^2 Law; Chen et al., 2025), recovery data selection (PASER; He et al., 2025), joint pruning-tuning (PAT; Liu et al., 2025b), post-hoc attention compensation (RestoreLCC; Feng et al., 2025), iterative prune-tune loops (Zhao et al., 2026), post-RLVR compression (OPSD; Kim and Lee, 2026), and rollout pruning during RLVR (Xu et al., 2026); none study which *pruning criterion* enables RL-based recovery. The Lottery Ticket Hypothesis (Frankle

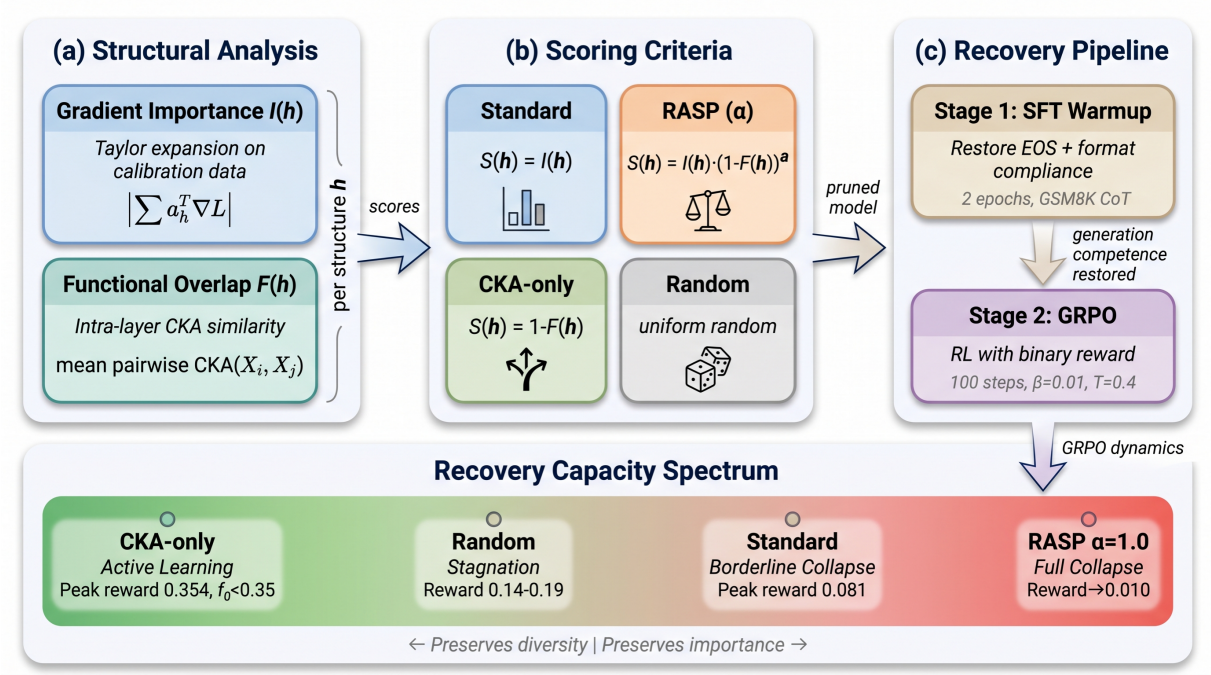


Figure 2: Overview of our pruning comparison framework. **Left**: structural analysis computes importance $I(h)$ and CKA-based overlap $F(h)$ scores, which four scoring criteria combine differently (Eqs. 4–6). **Center**: ranked structures are removed up to the target sparsity, subject to per-layer capacity floors. **Right**: all pruned models undergo identical two-stage recovery (SFT warmup → GRPO). **Bottom**: the resulting recovery capacity spectrum, from active learning (CKA-only) to training failure (RASP at α=1.0), tracks preserved representational diversity rather than gradient importance.

and Carbin, 2019) asks which subnetworks train effectively from scratch; our recovery capacity spectrum poses the analogous question for post-pruning recovery. While LTH concerns pruning at initialization seeking sparse subnetworks that train from scratch, our setting involves post-training structured pruning followed by recovery; the analogy is conceptual (pruning strategy affects trainability) rather than technical. Concurrent work benchmarks reasoning model compression (Zhang et al., 2025) and studies whether pruning can improve reasoning efficiency (Zhao et al., 2025), but neither isolates the pruning criterion's effect on RL recovery.

RLVR training dynamics. Matsutani et al. (2026) show that RLVR *compresses* incorrect reasoning trajectories while SFT *expands* correct ones, predicting RL–SFT complementarity. Our results qualify this prediction: the asymmetry manifests as a recovery capacity difference (the spectrum), but the expected accuracy gap is not observed when pruning preserves representational diversity ($|\Delta| \leq 1.8$ pp across GSM8K, MBPP, and MATH-500; §4.3). RL instability is well studied in the RL from Human Feedback (RLHF) context (Gao et al., 2023; Dong et al., 2025; Shah et al., 2025); Mukherjee et al. (2025) find that RL fine-tuning updates only 5–30% of parameters, which raises the question of whether pruning criteria interact with this active subnetwork. We connect these observations to structural pruning: the pruning criterion appears to shape susceptibility to RL collapse, while accuracy remains largely insensitive to the recovery paradigm at 20–25% sparsity within our experimental scope.

3 Method

To isolate how the pruning criterion affects post-RLVR recovery, we hold everything else constant: a shared structural analysis computes importance and functional overlap scores for all prunable structures, four scoring criteria combine these scores differently to select structures for removal, and an identical two-stage recovery pipeline is applied to every pruned model. The only variable across conditions is which structures are removed (Figure 2).

3.1 Preliminaries and Notation

Structured pruning. Given a pretrained transformer $\mathcal{M}$ with L layers, structured pruning removes entire computational units (attention heads

or MLP channels) to reduce model size while maintaining hardware-friendly dense computation. We denote the set of prunable structures as $\mathcal{H} = \{h_1, h_2, \ldots, h_N\}$. In this work, we focus exclusively on MLP channel group pruning; the formalism below covers attention heads for completeness, but all experiments remove only MLP groups.

Grouped Query Attention (GQA). In GQA architectures (Ainslie et al., 2023), n_Q query heads share n_{KV} key-value heads ($G = n_Q/n_{KV}$ per group). For DeepSeek-R1-Distill-Qwen-7B (n_Q=28, n_{KV}=4, G=7), we treat each GQA group (7 Q heads + 1 KV head) as a single prunable unit.

RLVR and GRPO. Reinforcement Learning with Verifiable Rewards (RLVR) recovers pruned model capabilities using outcome-based rewards from tasks with verifiable answers. We use Group Relative Policy Optimization (GRPO; Shao et al., 2024) as the recovery algorithm.

3.2 Structural Analysis

All four pruning criteria share a common structural analysis stage that computes two scores per structure: *importance* $I(h)$ capturing the structure's contribution to reasoning, and *functional overlap* $F(h)$ measuring representational redundancy with other structures in the same layer.

Calibration data. We compute importance and overlap scores on chain-of-thought reasoning data from GSM8K (Cobbe et al., 2021) training examples. Given a calibration set $\mathcal{D}_{\text{cal}} = \{(x_j, y_j)\}_{j=1}^{M}$ with M=512 randomly sampled examples, we compute activations and gradients on the concatenated sequences (x_j, y_j).

Structural importance. For each structure h with activation $\mathbf{a}_h$, we approximate the loss change from removal via first-order Taylor expansion ($\Delta\mathcal{L} \approx -\mathbf{a}_h^\top \nabla_{\mathbf{a}_h}\mathcal{L}$, dropping $O(\|\mathbf{a}_h\|^2)$ terms) and aggregate over the calibration set:

$$I(h) = \left| \sum_{(x,y)\in\mathcal{D}_{\text{cal}}} \mathbf{a}_h(x,y)^\top \frac{\partial \mathcal{L}(x,y)}{\partial \mathbf{a}_h} \right|, \quad (1)$$

where $\mathcal{L}$ is the language modeling loss over the reasoning trace. For a GQA group g containing Q heads $\{q_1, \ldots, q_G\}$ and KV head k, we aggregate: $I(g) = \sum_{i=1}^{G} I(q_i) + I(k)$.

Functional overlap via CKA. We measure functional overlap using linear Centered Kernel Alignment (CKA; Kornblith et al., 2019), which quantifies similarity between two sets of representations invariant to orthogonal transformations and isotropic scaling. Given activation matrices $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{n\times d}$ collected over n calibration examples, linear CKA is:

$$\text{CKA}(\mathbf{X}, \mathbf{Y}) = \frac{\text{HSIC}(\mathbf{X}, \mathbf{Y})}{\sqrt{\text{HSIC}(\mathbf{X}, \mathbf{X}) \cdot \text{HSIC}(\mathbf{Y}, \mathbf{Y})}}, \quad (2)$$

where HSIC is the Hilbert-Schmidt Independence Criterion, $\text{HSIC}(\mathbf{X}, \mathbf{Y}) = \frac{1}{(n-1)^2}\,\text{tr}(\mathbf{KHLH})$, with $\mathbf{K} = \mathbf{X}\mathbf{X}^\top$, $\mathbf{L} = \mathbf{Y}\mathbf{Y}^\top$, and $\mathbf{H} = \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$.

For a GQA group g_i in layer ℓ, we form its representation by concatenating its constituent Q head activations: $\mathbf{X}_{g_i} = [\mathbf{a}_{q_1}; \ldots; \mathbf{a}_{q_G}] \in \mathbb{R}^{n\times Gd_h}$. The functional overlap of g_i is the mean pairwise CKA with other groups in the same layer:

$$F(g_i) = \frac{1}{|\mathcal{G}_\ell| - 1} \sum_{\substack{g_j \in \mathcal{G}_\ell \\ j \neq i}} \text{CKA}(\mathbf{X}_{g_i}, \mathbf{X}_{g_j}). \quad (3)$$

We apply the same intra-layer CKA overlap to MLP channel groups, partitioning each MLP intermediate dimension into equal-sized groups. Unlike prior work that uses inter-layer CKA to identify globally redundant layers for depth pruning (Hu et al., 2024; Lachance, 2022), we use *intra-layer* CKA between individual structures to measure per-structure functional redundancy. Computing all pairwise CKA scores adds minimal overhead ($O(L \cdot k^2)$, $k \leq 28$; <2 min on one GPU).

CKA reliability. Our intra-layer use between structures of identical dimensionality avoids sensitivities identified for cross-layer comparisons (Davari et al., 2023; Klabunde et al., 2023); rankings are stable across calibration sizes (Spearman $\rho \geq 0.964$; Appendix H).

3.3 Pruning Scoring Criteria

We compare four scoring criteria that combine importance $I(h)$ and functional overlap $F(h)$ in different ways. All four methods use the same per-layer capacity floor (≥40% of MLP groups per layer must be retained, chosen to prevent any single layer from becoming a throughput bottleneck; ablation over $\tau \in \{0.3, 0.4, 0.5, 0.6\}$ in Appendix S) and global ranking procedure: structures

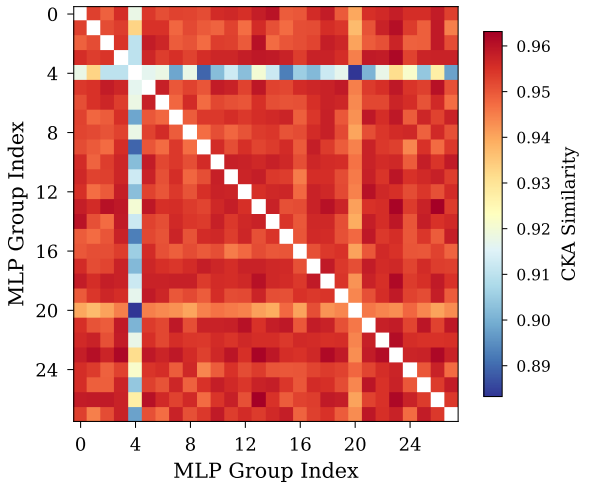


Figure 3: Intra-layer pairwise CKA similarity between 28 MLP channel groups in Layer 27 of DeepSeek-R1-Distill-Qwen-7B. Most groups are highly redundant (CKA $>$ 0.93, red), but Group 4 exhibits distinctly lower similarity (blue row/column), indicating a functionally unique computational pathway. CKA-only scoring (Eq. 6) preserves such unique structures while removing the redundant majority.

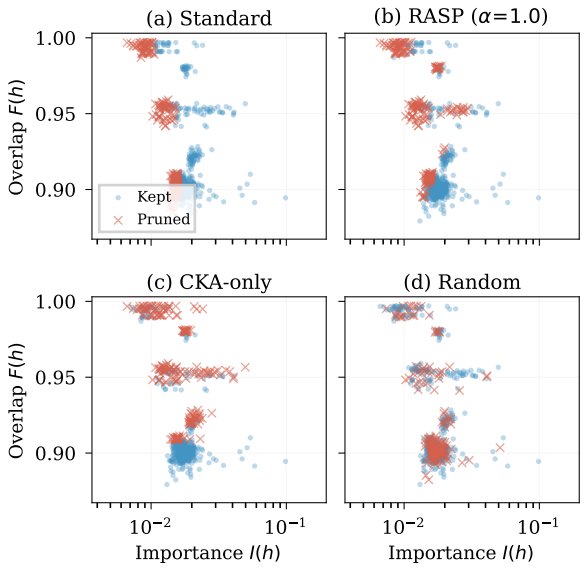


Figure 4: Pruning decisions of four criteria at 20% MLP sparsity. Each point is one MLP channel group; $\times$ (red) = pruned, $\circ$ (blue) = retained. (a) Standard prunes low-importance structures regardless of overlap. (b) RASP at α=1.0 targets structures with both low importance and high overlap. (c) CKA-only prunes high-overlap structures regardless of importance. (d) Random selects uniformly.

are sorted by score, and the lowest-scoring structures are pruned up to the target sparsity ratio, skipping any candidate whose removal would violate the layer floor.

Standard (importance-only). Following LLM-Pruner (Ma et al., 2023), structures are ranked solely by importance:

$$S_{\text{std}}(h) = I(h). \tag{4}$$

This is the industry-standard criterion that retains gradient-important structures and removes "unimportant" ones.

RASP (recovery-aware composite). RASP is a parameterized family of scoring functions indexed by α, which controls the trade-off between gradient importance and diversity preservation:

$$S_{\text{rasp}}(h) = I(h) \times \big(1 - F(h)\big)^{\alpha}, \tag{5}$$

where α=0 reduces to Standard and larger α increasingly favors diversity. The multiplicative form ensures zero-importance structures are always pruned (S=0) regardless of overlap. Dose-response analysis (§4.4) reveals an inverted-U peaking at α=2.0 (66.11%), where the diversity term $(1-F(h))^2$ dominates but importance filtering still provides complementary signal (+1.72 pp over CKA-only; Table 5).

CKA-only (overlap-only). Structures are ranked by functional overlap alone, removing those with the highest mean pairwise CKA first:

$$S_{\text{cka}}(h) = 1 - F(h), \tag{6}$$

corresponding to $\alpha \to \infty$ in Eq. 5. This preserves all representationally unique structures regardless of gradient importance.

Random. Structures are selected uniformly at random for removal, subject to the per-layer capacity floor. This baseline controls for the effect of any informed scoring criterion.

The four criteria span a spectrum from pure importance (Standard) to pure diversity preservation (CKA-only), with RASP as a parameterized interpolation (Figure 4). CKA-only retains *functionally unique* structures regardless of gradient contribution, preserving non-redundant computational pathways that may support RL exploration (Matsutani et al., 2026).

3.4 Two-Stage Recovery

After pruning, all four models undergo identical two-stage recovery: (1) a brief SFT warmup on chain-of-thought demonstrations restores end-of-sequence generation and format compliance (prun-

Table 1: Recovery capacity spectrum at 20% sparsity. f_0: zero-variance batch fraction. CKA-only sustains active learning; Random stagnates; Standard shows borderline collapse; RASP exhibits full collapse. Best in **bold**.

Method	Peak Rew. ↑	Mean Rew. ↑	Min f_0 ↓
CKA-only	**0.354**	**0.285**	**0.125**
Random	0.190	0.164	0.300
Standard	0.081	0.050	0.500
RASP (α=1.0)	0.061	0.031	0.675

ing disrupts termination behavior, so direct GRPO is infeasible); (2) GRPO with binary reward enables RL-based exploration and redistributes functionality across remaining structures (Matsutani et al., 2026), in contrast to SFT which imitates fixed trajectories. Details in Section 4.1 and Appendix A.

3.5 Recovery Capacity Diagnostics

We define **recovery capacity** as peak mean reward over the GRPO training horizon, with four diagnostics (reward, zero-variance batch fraction f_0, gradient norm, KL divergence) distinguishing *active learning*, *stagnation*, and *collapse* regimes (Algorithm 1, Appendix B).

4 Experiments

4.1 Experimental Setup

Base model and evaluation. We evaluate on DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025), a 7B-parameter reasoning model built on the Qwen2.5 architecture (Yang et al., 2024a) with GQA (Ainslie et al., 2023) (Section 3.1). Our primary benchmark is GSM8K (Cobbe et al., 2021) (N=1,319 grade-school math problems; exact-match accuracy). The unpruned model achieves 90.83% (greedy decoding).

Pruning methods. We compare four pruning criteria at 20% structured sparsity, all removing 200 of 784 MLP channel groups across 28 layers (∼1.45B parameters) with identical per-layer capacity floors (≥40% retention): **CKA-only** (Eq. 6), **Standard** (Eq. 4; LLM-Pruner (Ma et al., 2023)), **RASP** at α=1.0 (Eq. 5), and **Random** (uniform selection; uses TRL's default 512-token generation limit vs. 1,024 for other methods). See Section 3.3 and Figure 4 for scoring details.

Recovery pipeline and hardware. Immediately after pruning, all four methods produce near-zero GSM8K accuracy (<2%). All methods undergo matched two-stage recovery (Section 3.4): SFT warmup (2 epochs, 7,473 GSM8K examples, 37 min) followed by GRPO (lr 1×10^{-6}, β=0.01, T=0.4, 100 steps on 1,000 problems; full hyperparameters in Table 3, Appendix A). We repeat GRPO with 3 seeds for CKA-only, Standard, and Random (RASP at α=1.0 uses a single seed); CKA-only achieves 64.39% ± 0.19 pp, Standard 59.84% ± 0.38 pp (Welch's t, p<0.001; Appendix J). The "Unpruned" baseline in Table 2 uses the same SFT-warmup checkpoint; the 9.56 pp gap from the pre-warmup 90.83% reflects warmup on a reduced training set, not pruning loss. All experiments run on 2× NVIDIA RTX PRO 6000 GPUs (94 GB VRAM each).

4.2 Recovery Capacity Spectrum

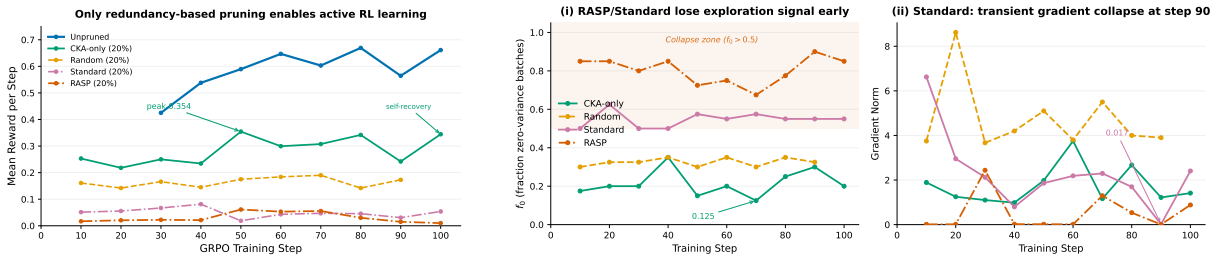
Under GRPO training, the four pruning criteria produce a clear recovery capacity spectrum: CKA-only ≫ Random ≫ Standard > RASP at α=1.0 (Table 1, Figure 5). The pruning criterion, rather than training configuration, determines whether the model learns, stagnates, or collapses.

Among these configurations, only the diversity-maximizing criterion (CKA-only) sustains active learning: peak reward 0.354 with f_0 as low as 0.125 (Figure 5b). Random, under a shorter generation limit, stagnates (reward 0.14–0.19, min $f_0 = 0.300$); Standard exhibits borderline collapse ($f_0 \geq 0.50$, peak reward 0.081); RASP at α=1.0 exhibits complete reward collapse ($f_0 \geq 0.675$, reward declining to 0.010, with near-zero KL divergence consistent with minimal policy movement; Appendix Q).

This ordering does not mirror post-SFT accuracy. Under SFT alone the ranking is CKA-only > RASP at α=1.0 > Standard > Random, yet the spectrum reverses the RASP/Standard ordering: Standard exhibits higher peak reward than RASP despite lower SFT accuracy (Table 1). The spectrum thus tracks structural properties rather than training-stage accuracy, and the dose-response analysis (§4.4) confirms CKA weighting as the critical variable. Given the scale of these dynamics, one would expect equally large accuracy differences.

4.3 Reward-Accuracy Decoupling

The opposite holds. Despite the 4.4× peak reward gap, GRPO produces minimal accuracy change over SFT (Figure 6): at 20% sparsity, all fine-grained methods differ by $|\Delta| < 1$ pp (CKA-only



(a) Reward trajectories

(b) f_0 and gradient norm diagnostics

Figure 5: GRPO training dynamics for four pruning criteria (100 steps, β=0.01, T=0.4). (a) Reward trajectories: CKA-only reaches 0.354 peak; Random oscillates (0.14–0.19); Standard peaks at 0.081; RASP at α=1.0 declines to 0.010. (b) Left: f_0 (zero-variance batch fraction); CKA-only remains below 0.35 while Standard and RASP at α=1.0 persist in the collapse zone ($f_0 > 0.5$). Right: gradient norm; RASP at α=1.0 exhibits near-zero norms throughout; Standard shows a transient minimum at step 90.

Table 2: Reward-accuracy decoupling on GSM8K (N=1,319, greedy decoding) at step 100. All pruned models at 20% sparsity. Multi-seed (mean ± std) where available; single-seed entries lack ±. Best in **bold**. Peak Rew. not recorded for α ablation variants (–).

Method	SFT (%)	GRPO (%)	Δ (pp)	Peak Rew.
Unpruned	**81.27**	**81.27**	0.00	**0.669**
RASP (α=2.0)	64.97	66.11	+1.14	–
CKA-only	63.91	64.39±0.19	+0.48	0.354
Standard	59.29	59.84±0.38	+0.55	0.081
RASP (α=1.0)	61.49	60.88	−0.61	0.061
Random	51.25	50.72±0.15	−0.53	0.190
ShortGPT	44.73	43.37	−1.36	–

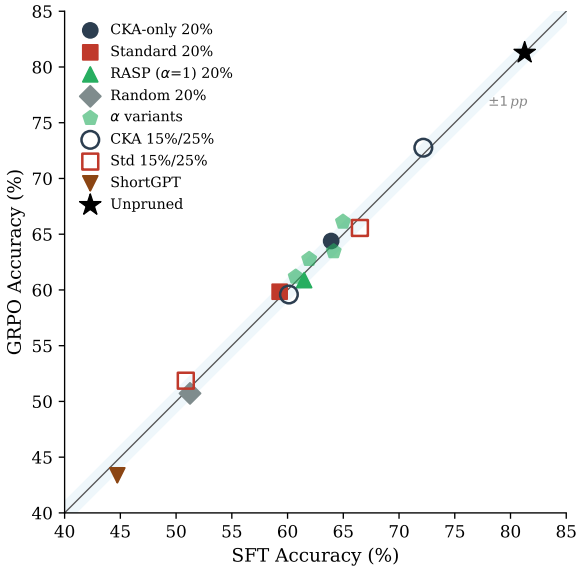


Figure 6: SFT vs. GRPO accuracy across all pruning criteria and sparsity levels on GSM8K (N=1,319). Points near the y=x line indicate reward-accuracy decoupling: despite 4.4× peak reward gaps (Figure 5a), GRPO yields minimal accuracy change ($|\Delta| \leq 1$ pp for all fine-grained methods at 20–25%). RASP points correspond to α=1.0.

+0.48, Standard +0.55, Random −0.53, RASP at α=1.0 −0.61 pp; Table 2). We term this *reward-accuracy decoupling* (observed at 20–25% fine-grained MLP sparsity): reward trajectory reflects the model's capacity to learn from RL signal, not downstream accuracy.

Decoupling is not universal: at 15% sparsity, GRPO amplifies the CKA-only advantage (SFT gap 5.69 pp → GRPO gap 7.20 pp), but at 20–25% $|\Delta|$ remains below 1 pp without exception (Tables 2, 12; Appendix M), and dose-response α variations stay within 1.14 pp (Table 5; Appendix F).

Small $|\Delta|$ alone does not establish equivalence; it could reflect insufficient power. Two One-Sided Tests (TOST; ε=2 pp ≈4× cross-seed σ; N=1,319 paired examples) confirm statistical equivalence for 7/8 comparisons (p<0.05; Appendix T), robust across seeds (0.37 pp range; Appendix L) and extended training (300 steps; §4.5). Directional trends hold on MATH-500 (Hendrycks et al., 2021) (CKA-only and Standard rank above RASP and Random; gap compression at higher difficulty; Appendix F) and MBPP (Austin et al., 2021) (all fine-grained GRPO $|\Delta| < 2$ pp; Appendix G). ShortGPT (Men et al., 2024) depth pruning is the sole exception: its GRPO delta is task-dependent (−1.36 pp on GSM8K, +4.01 pp on MBPP; Tables 2, 7), unlike fine-grained methods ($|\Delta| \leq$ 1.8 pp), which suggests that layer-level removal interacts differently with RL recovery.

4.4 Dose-Response Analysis

The recovery capacity spectrum and decoupling raise a natural question: what controls the balance between RL trainability and final accuracy? The CKA coefficient α provides a continuous knob. On

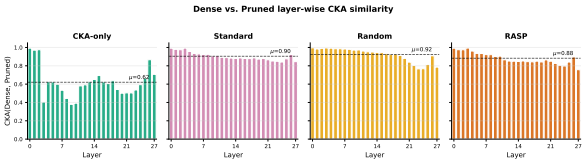


Figure 7: Layer-wise CKA similarity between dense and pruned representations (28 layers). CKA-only diverges most from the dense model (μ=0.62) yet achieves the highest recovery; Standard (μ=0.90), Random (μ=0.92), and RASP at α=1.0 (μ=0.88) preserve dense fidelity but exhibit reward collapse under RL.

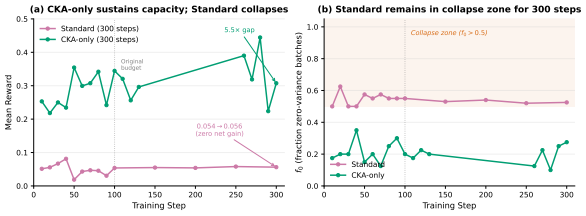


Figure 8: Extended GRPO training (300 steps, 3× budget). (a) Standard reward: zero net gain (0.054→0.056); CKA-only sustains a 5.5× gap. (b) Standard f_0 remains in the collapse zone (>0.5); CKA-only stays below 0.30.

GSM8K, recovery follows an inverted-U across $\alpha \in \{0.5, 1.0, 2.0, 3.0, 4.0\}$, peaking at α=2.0 (66.11%; Figure 1, Table 5) with $|\Delta| \leq 1.14$ pp; RASP at α=1.0 exhibits reward collapse while RASP at α=4.0 recovers to 62.77% (Appendix D). The pattern generalizes to MATH-500 (α=1.0 worst at 23.0%; higher-α cluster at 27.2–27.6%; Table 6). All methods collapse at 30% sparsity (Appendix K); pruning reduces parameters by 19% and memory by ∼28% (Appendix C).

4.5 Structural Evidence

Representational diversity. If diversity preservation drives recovery capacity, CKA-only should diverge most from the dense model in representation space. Figure 7 confirms this: CKA-only shows the lowest layer-wise CKA similarity (μ=0.62), while Standard (μ=0.90), Random (μ=0.92), and RASP at α=1.0 (μ=0.88) cluster near 1.0; high fidelity to the dense model comes at the cost of diversity. CKA-only concentrates pruning in early layers (L0–9, 72.5% of removed groups; Appendix P, R), yielding 2.89× mean gradient magnitude versus Standard (55/56 layer groups p<0.05; Appendix I). On MATH-500, CKA-only truncates only 22% of completions (mean 655 tokens) versus Standard's 44% (mean 1,128 tokens), yet achieves equal or higher accuracy (Table 6).

Extended training does not resolve collapse. A natural objection is that importance-dominated criteria simply need more training. When GRPO is extended from 100 to 300 steps (Figure 8; Appendix E), Standard shows no net improvement (reward 0.054–0.058, $f_0 > 0.5$ throughout), while CKA-only sustains a 5.5× gap (0.308 vs. 0.056 at step 300). This persistent gap confirms that the divergence is structural rather than a matter of training budget.

Cross-scale and cross-model generalization. The spectrum ordering also holds beyond the primary model. Preliminary evidence at 1.5B scale (Table 13; Appendix N) preserves the three-method ordering, and on Qwen2.5-Math-7B (Appendix O), CKA-only recovers from 1.0% post-prune to 51.50% vs. Standard's 21.00%.

5 Conclusion

Whether RLVR outperforms SFT for recovering pruned reasoning models appears to depend on how the model was pruned. Under diversity-preserving pruning at 20–25% MLP sparsity on Qwen2.5-family models, the two paradigms produce near-identical accuracy (mean $|\Delta|$=0.54 pp; max 1.8 pp), yet the pruning criterion creates a 4.4× peak reward gap. SFT warmup recovers the bulk of accuracy (from <2% post-prune to 51–64%); GRPO exposes structural capacity differences but contributes minimal additional accuracy, a pattern that persists at 300 steps (Appendix E). The non-monotonic α dose-response argues against a layer-distribution confound. A practical implication is that GRPO reward trajectories and f_0 diagnostics can serve as probes for evaluating pruning quality, surfacing structural deficiencies that accuracy alone would miss. We discuss broader implications in Appendix U.

Limitations

Our primary results are on DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025) with GQA (G=7); cross-model validation on Qwen2.5-Math-7B covers SFT recovery only (N=200), and both models share the Qwen2.5 base architecture, so generalization to different architecture families (e.g., LLaMA, Mistral) and scales (1B–70B) remains open. Reward-accuracy decoupling at 20–25% sparsity means that recovery capacity does not predict downstream accuracy improvements within

our training horizon; practitioners seeking accuracy gains may require longer training, different reward formulations, or lower sparsity levels. At 20% sparsity, pruning reduces memory by ∼28% but inference throughput decreases by 19% (72.0→58.3 tok/s), likely because channel-group pruning creates irregular patterns that current hardware cannot accelerate. CKA-only, Standard, and Random are validated over 3 GRPO seeds while RASP at α=1.0 uses a single seed; Random uses a 512-token generation limit (vs. 1,024 for other methods), so its ranking should be interpreted as a conservative bound. We evaluate on three benchmarks (GSM8K, MATH-500, MBPP) covering math reasoning and code generation; logical, commonsense, and multimodal reasoning are not addressed.

Ethics Statement

This work uses publicly available models (DeepSeek-R1-Distill-Qwen, Qwen2.5-Math) and established benchmarks (GSM8K, MATH-500, MBPP); no private data or human subjects are involved. Structured pruning reduces model size and inference memory, which can lower the computational cost and energy consumption of deploying large language models. Our research focuses on the technical analysis of pruning criteria and their interaction with reinforcement learning recovery; we do not foresee direct negative societal impacts from this work. All models and datasets used in this work are publicly available under permissive licenses: DeepSeek-R1-Distill-Qwen (MIT), Qwen2.5-Math (Apache 2.0), GSM8K (MIT), MATH (MIT), and MBPP (CC-BY-4.0). We disclose our use of AI assistants in Appendix V.

References

Joshua Ainslie, James Lee-Thorp, Michiel de Jong, Yury Zemlyanskiy, Federico Lebrón, and Sumit Sanghai. 2023. GQA: Training generalized multi-query transformer models from multi-head checkpoints. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Saleh Ashkboos, Maximilian L. Croci, Marcelo Gennari do Nascimento, Torsten Hoefler, and James Hensman. 2024. SliceGPT: Compress large language models by deleting rows and columns. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*.

Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, and Charles Sutton. 2021. Program synthesis with large language models. *arXiv preprint arXiv:2108.07732*.

Xiaodong Chen, Yuxuan Hu, Xiaokang Zhang, Yanling Wang, Cuiping Li, Hong Chen, and Jing Zhang. 2025. P^2 law: Scaling law for post-training after model pruning. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*.

MohammadReza Davari, Stefan Horoi, Amine Natik, Guillaume Lajoie, Guy Wolf, and Eugene Belilovsky. 2023. Reliability of CKA as a similarity measure in deep learning. In *Proceedings of the 11th International Conference on Learning Representations (ICLR)*.

DeepSeek-AI. 2025. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*.

Longwei Ding, Anhao Zhao, Fanghua Ye, Ziyang Chen, and Xiaoyu Shen. 2026. From LLMs to LRMs: Rethinking pruning for reasoning-centric models. *arXiv preprint arXiv:2601.18091*.

Yihong Dong, Xue Jiang, Yongding Tao, Huanyu Liu, Kechi Zhang, Lili Mou, Rongyu Cao, Yingwei Ma, Jue Chen, Binhua Li, Zhi Jin, Fei Huang, Yongbin Li, and Ge Li. 2025. RL-PLUS: Countering capability boundary collapse of LLMs in reinforcement learning with hybrid-policy optimization. *arXiv preprint arXiv:2508.00222*.

Zijian Feng, Hanzhang Zhou, Zixiao Zhu, Tianjiao Li, Jia Jim Deryl Chua, Lee Onn Mak, Gee Wah Ng, and Kezhi Mao. 2025. RestoreLCC: Restoring pruned large language models via lost component compensation. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Jonathan Frankle and Michael Carbin. 2019. The lottery ticket hypothesis: Finding sparse, trainable neural networks. In *Proceedings of the 7th International Conference on Learning Representations (ICLR)*.

Leo Gao, John Schulman, and Jacob Hilton. 2023. Scaling laws for reward model overoptimization. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*.

Bowei He, Lihao Yin, Hui-Ling Zhen, Xiaokun Zhang, Mingxuan Yuan, and Chen Ma. 2025. Paser: Post-training data selection for efficient pruned large language model recovery. *arXiv preprint arXiv:2502.12594*. Under review.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. 2021. Measuring mathematical problem solving with the MATH dataset. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Seungbeom Hu, Chanjun Park, Andrew Ferraiuolo, Sang-Ki Ko, Jinwoo Kim, Haein Song, and Jieung Kim. 2024. Mpruner: Optimizing neural network size with cka-based mutual information pruning. *arXiv preprint arXiv:2408.13482*.

Yixin Ji, Zhimin Lin, Juntao Li, Kaixin Wu, Kai Yao, Xu Jia, Linjian Mo, and Min Zhang. 2025. Self-healing: Recovering pruned large reasoning models via reinforcement learning. *OpenReview preprint*.

Jaehoon Kim and Dongha Lee. 2026. OPSD compresses what RLVR teaches: A post-RL compaction stage for reasoning models. *arXiv preprint arXiv:2605.06188*.

Max Klabunde, Tobias Schumacher, Markus Strohmaier, and Florian Lemmerich. 2023. Similarity of neural network models: A survey of functional and representational measures. *ACM Computing Surveys*, 57(9).

Steven Kolawole, Lucio Dery, Jean-François Kagy, Virginia Smith, Graham Neubig, and Ameet Talwalkar. 2024. Everybody prune now: Structured pruning of llms with only forward passes. *arXiv preprint arXiv:2402.05406*.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. 2019. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*.

Alex Lachance. 2022. Using centered kernel alignment for understanding and pruning neural networks. Master's thesis, University of Rhode Island.

Yiheng Liu, Junhao Ning, Sichen Xia, Xiaohui Gao, Ning Qiang, Bao Ge, Junwei Han, and Xintao Hu. 2025a. Pruning large language models by identifying and preserving functional networks. *arXiv preprint arXiv:2508.05239*.

Yijiang Liu, Huanrui Yang, Youxin Chen, Rongyu Zhang, Miao Wang, Yuan Du, and Li Du. 2025b. PAT: Pruning-aware tuning for large language models. In *Proceedings of the 39th AAAI Conference on Artificial Intelligence (AAAI)*.

Ryan Lucas, Kayhan Behdin, Zhipeng Wang, Qingquan Song, Shao Tang, and Rahul Mazumder. 2025. Reasoning models can be accurately pruned via chain-of-thought reconstruction. In *NeurIPS 2025 Workshop on Efficient Reasoning*.

Xinyin Ma, Gongfan Fang, and Xinchao Wang. 2023. Llm-pruner: On the structural pruning of large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Kohsei Matsutani, Shota Takashiro, Gouki Minegishi, Takeshi Kojima, Yusuke Iwasawa, and Yutaka Matsuo. 2026. Rl squeezes, sft expands: A comparative study of reasoning llms. In *Proceedings of the 14th International Conference on Learning Representations (ICLR)*.

Xin Men, Mingyu Xu, Qingyu Zhang, Bingning Wang, Hongyu Lin, Yaojie Lu, Xianpei Han, and Weipeng Chen. 2024. ShortGPT: Layers in large language models are more redundant than you expect. *arXiv preprint arXiv:2403.03853*.

Ocean Monjur, Shahriar Kabir Nahin, and Anshuman Chhabra. 2026. Doing more with less: Revisiting the effectiveness of LLM pruning for test-time scaling. *arXiv preprint arXiv:2604.25098*.

Sagnik Mukherjee, Lifan Yuan, Dilek Hakkani-Tür, and Hao Peng. 2025. Reinforcement learning finetunes small subnetworks in large language models. *arXiv preprint arXiv:2505.11711*.

Vedant Shah, Johan Obando-Ceron, Vineet Jain, Brian Bartoldson, Bhavya Kailkhura, Sarthak Mittal, Glen Berseth, Pablo Samuel Castro, Yoshua Bengio, Nikolay Malkin, Moksh Jain, Siddarth Venkatraman, and Aaron Courville. 2025. A comedy of estimators: On KL regularization in RL training of LLMs. *arXiv preprint arXiv:2512.21852*.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Mingchuan Zhang, Y.K. Li, Y. Wu, and Daya Guo. 2024. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*.

Safal Shrestha, Anubhav Shrestha, Aadim Nepal, Minwu Kim, and Keith Ross. 2026. On the limits of layer pruning for generative reasoning in LLMs. *arXiv preprint arXiv:2602.01997*.

Ziyan Wang, Enmao Diao, Qi Le, Pu Wang, Guanchu Wang, Minwoo Lee, Shu-ping Yeh, and Li Yang. 2025. Think before you prune: Self-reflective structured pruning for reasoning language models. *arXiv preprint arXiv:2512.02185*.

Ziyang Wang, Jiangfeng Xiao, Chuan Xiao, Ruoxiang Li, Rui Mao, and Jianbin Qin. 2026. Grasprune: Global gating for budgeted structured pruning of large language models. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*.

Yang Xiang, Yixin Ji, Juntao Li, and Min Zhang. 2025. Think before you prune: Selective self-generated calibration for pruning large reasoning models. *arXiv preprint arXiv:2511.18864*.

Haobo Xu, Sirui Chen, Ruizhong Qiu, Yuchen Yan, Chen Luo, Monica Cheng, Jingrui He, and Hanghang Tong. 2026. Prune as you generate: Online rollout pruning for faster and better RLVR. *arXiv preprint arXiv:2603.24840*.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, and 1 others. 2024a. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Yifei Yang, Zouying Cao, and Hai Zhao. 2024b. LaCo: Large language model pruning via layer collapse. In *Findings of the Association for Computational Linguistics: EMNLP 2024*.

Tao Yu, Yongqi An, Kuan Zhu, Guibo Zhu, Ming Tang, and Jinqiao Wang. 2025. FANG: Improving generalization in LLM structured pruning via function-aware neuron grouping. *arXiv preprint arXiv:2512.23014*.

Nan Zhang, Eugene Kwek, Yusen Zhang, Ngoc-Hieu Nguyen, Prasenjit Mitra, and Rui Zhang. 2025. When reasoning meets compression: Understanding the effects of LLMs compression on large reasoning models. *arXiv preprint arXiv:2504.02010*.

Shangziqi Zhao, Jiahao Yuan, Jinyang Wu, Zhenglin Wang, Guisong Yang, and Usman Naseem. 2025. Can pruning improve reasoning? revisiting long-CoT compression with capability in mind for better reasoning. *arXiv preprint arXiv:2505.14582*.

Yiran Zhao, Shengyang Zhou, Zijian Wu, Tongyan Hu, Yuhui Xu, Rengan Dou, Kenji Kawaguchi, Shafiq Joty, Junnan Li, and Michael Qizhe Shieh. 2026. Gradually compacting large language models for reasoning like a boiling frog. *arXiv preprint arXiv:2602.04919*.

A Training Details

Table 3 lists the full hyperparameter configurations for both recovery stages described in Section 4.1. All four pruning criteria share identical SFT and GRPO settings; the only difference is Random's shorter generation limit, inherited from the TRL library default.

Table 3: Hyperparameter details for SFT warmup and GRPO recovery stages. All methods share the same GRPO configuration; Random uses the TRL default generation limit (marked with †).

Parameter	SFT	GRPO
Learning rate	2×10^{-5}	1×10^{-6}
Batch size	1	1
Gradient accumulation	8	1
Max sequence length	2,048	–
Max new tokens	–	1,024 †512
Optimizer	AdamW	AdamW
Epochs / Steps	2 epochs	100 steps
KL coefficient β	–	0.01
Sampling temperature	–	0.4
Num generations per prompt	–	8
Reward	–	Binary (exact match)
Loss formulation	Cross-entropy	DAPO-style clipped
`reward_desert_patience`	–	50

†Random uses TRL default of 512 tokens; all other methods use 1,024.

B Algorithm

Algorithm 1 summarizes the complete pruning comparison procedure from Section 3: shared structural analysis (importance and CKA overlap computation), criterion-specific scoring (Eqs. 4–6), and two-stage recovery (SFT warmup followed by GRPO).

Algorithm 1 Controlled Pruning Comparison Framework

Require: Model $\mathcal{M}$, calibration data $\mathcal{D}_{\text{cal}}$, target sparsity s, scoring criterion $\mathcal{C} \in \{\text{Std}, \text{RASP}, \text{CKA}, \text{Rand}\}$, layer floor τ
Ensure: Pruned and recovered model $\mathcal{M}'$

% Stage 1: Shared Structural Analysis
for each structure $h \in \mathcal{H}$ **do**
  Compute importance $I(h)$ via Eq. (1)
  Collect activations $\mathbf{X}_h$ on $\mathcal{D}_{\text{cal}}$
for each layer $\ell = 1, \ldots, L$ **do**
  for each structure h_i in layer ℓ **do**
    $F(h_i) \leftarrow \frac{1}{|\mathcal{H}_\ell|-1} \sum_{j \neq i} \text{CKA}(\mathbf{X}_{h_i}, \mathbf{X}_{h_j})$
% Stage 2: Criterion-Specific Scoring
for each structure $h \in \mathcal{H}$ **do**
  $S(h) \leftarrow \mathcal{C}(I(h), F(h))$ ▷ Eq. 4, 5, 6, or random
Sort $\mathcal{H}$ by $S(h)$ ascending; prune lowest up to s, subject to τ
% Stage 3: Two-Stage Recovery
$\mathcal{M}_{\text{warm}} \leftarrow \text{SFT}(\mathcal{M}_{\text{pruned}}, \mathcal{D}_{\text{sft}})$ ▷ restore EOS & format
$\mathcal{M}' \leftarrow \text{GRPO}(\mathcal{M}_{\text{warm}}, \mathcal{D}_{\text{rlvr}})$ ▷ recover reasoning
return $\mathcal{M}'$

C Inference Efficiency

Table 4: Inference characteristics at 20% sparsity (single-GPU, no contention). Pruning reduces parameters by 19% and memory by ∼28%, with TTFT improving 18%. Per-token throughput decreases 19% (0.81×) due to non-uniform layer widths from full-layer removal that current GPU kernels do not optimize.

Configuration	TPS (tok/s)	TTFT (ms)	Per-token (ms)	Params (B)
Dense 7B	72.0	20.1	13.9	7.62
CKA-only 20%	58.3	16.4	16.7	6.16
Standard 20%	58.2	16.4	16.7	6.16

All pruned models achieve ∼28% memory reduction from structural parameter removal. TTFT (time to first token) improves by 18%

Table 5: Dose-response analysis: effect of CKA coefficient α on GSM8K accuracy (N=1,319) at 20% MLP channel sparsity. α=0 reduces to Standard (importance-only); higher α weights representational diversity. Best in **bold**.

α	SFT Acc. (%)	GRPO Acc. (%)	Δ (pp)
0.5	60.73	61.18	+0.45
1.0	61.49	60.88	−0.61
2.0	**64.97**	**66.11**	+1.14
3.0	64.14	63.46	−0.68
4.0	61.94	62.77	+0.83

(20.1→16.4 ms), reflecting faster prefill with fewer parameters. Per-token throughput decreases by 19% (72.0→58.3 tok/s, or 13.9→16.7 ms per token), consistent with non-uniform layer widths creating computational bottlenecks that current GPU kernels do not optimize away. CKA-only and Standard show identical throughput (58.3 vs 58.2 tok/s), confirming that the pruning criterion affects recovery capacity but not inference speed at matched sparsity.

D CKA Coefficient α Ablation

On GSM8K, the dose-response curve across $\alpha \in \{0.5, 1.0, 2.0, 3.0, 4.0\}$ reveals an inverted-U pattern peaking at α=2.0 (SFT 64.97%, GRPO 66.11%, Δ=+1.14 pp; Table 5). At α=0.5, importance still dominates; recovery is low (SFT 60.73%, GRPO 61.18%). At α=1.0 (RASP), GRPO collapses (peak reward 0.061, accuracy 60.88%, Δ=−0.61 pp). α=3.0 (SFT 64.14%, GRPO 63.46%, Δ=−0.68 pp) excludes a plateau hypothesis: accuracy drops 2.65 pp from the α=2.0 peak, confirming that the optimum is a true peak rather than a saturation regime. At α=4.0, CKA overlap dominates; the model recovers to 62.77% GRPO accuracy (SFT 61.94%, Δ=+0.83 pp) with active GRPO learning (reward 0.146→0.256→0.198). Reward-accuracy decoupling holds at every α value: $|\Delta| \leq 1.14$ pp across all five points.

E Extended Training Dynamics

To test whether importance-dominated criteria simply need more training to overcome collapse (§4.5), we extend GRPO from 100 to 300 steps for CKA-only and Standard (Figure 8). At 300 steps, GSM8K accuracy remains unchanged: CKA-only 64.37% (identical to the 100-step seed 42 result), Standard 59.29% (within the 100-step seed variance of ±0.38 pp). CKA-only reward rises from 0.354 (peak at step 100) to 0.308 (step 300), yet accuracy does not improve, directly confirming reward-accuracy decoupling at this training horizon. Standard reward remains flat (0.054–0.058) with $f_0 > 0.5$ throughout, confirming that additional training does not resolve the structural collapse identified in Section 4.2.

F MATH-500 Out-of-Distribution Evaluation

Table 6: MATH-500 out-of-distribution evaluation (N=500, greedy decoding, `max_new_tokens=2048`). CKA-only and Standard both outperform RASP and Random; the CKA-only–Standard GRPO gap is 0.6 pp (vs. 4.62 pp on GSM8K), consistent with difficulty and truncation compressing scores toward a floor. Best per section in **bold**.

Method	SFT (%)	GRPO (%)	Δ (pp)
Dense		53.0 †	
20% sparsity — method comparison			
CKA-only	**27.4**	**27.6**	+0.2
Standard	25.6	27.0	+1.4
RASP (α=1.0)	21.6	23.0	+1.4
Random	16.6	15.0	−1.6
20% sparsity — α dose-response			
α=0.5	26.8	26.8	+0.0
α=2.0	25.6	**27.4**	+1.8
α=3.0	23.6	–	–
α=4.0	**27.8**	27.2	−0.6
25% sparsity			
CKA-only	18.8	17.6	−1.2
Standard	15.4	–	–

† Over half of Dense responses truncated at 2048 tokens (73.6% at 4096); all pruned models evaluated under the same constraint.

CKA-only and Standard both outperform RASP and Random on MATH-500 (SFT: 27.4%, 25.6% vs. 21.6%, 16.6%; GRPO: 27.6%, 27.0% vs. 23.0%, 15.0%), consistent with GSM8K. However, the CKA-only–Standard gap narrows substantially: from 4.6 pp (GSM8K SFT) to 1.8 pp (MATH-500 SFT) to 0.6 pp (MATH-500 GRPO). Middle-ranked methods also swap positions (Standard > RASP on MATH-500 vs. RASP > Standard on GSM8K). This narrowing reflects two compounding factors: MATH-500's higher intrinsic difficulty (the unpruned model achieves only 53.0% vs. 90.83% on GSM8K) and generation-length truncation at the 2,048-token evaluation limit (over half of dense-model responses are truncated; Table 6), both of which compress all pruned methods toward a per-

formance floor and reduce inter-method differentiation (§4.3). CKA-only also produces more efficient reasoning: only 22% of completions are truncated at the 2,048-token limit (mean 655 tokens) versus Standard's 44% (mean 1,128 tokens), indicating that the preserved structure enables shorter reasoning chains rather than longer ones. The α dose-response effect is not statistically significant on MATH-500: SFT accuracy varies by $\sim$8 pp across α values, but with stderr $\approx$2 pp per point, attributable to high truncation rates at the 2048-token limit and floor effects at this difficulty level. The inverted-U pattern is validated on GSM8K (Table 5); MATH-500 does not resolve the dose-response shape.

G MBPP Code Generation

To test whether reward-accuracy decoupling extends beyond math reasoning (§4.3), we evaluate on MBPP (Austin et al., 2021) (N=374 code-generation problems; Table 7).

Table 7: MBPP code generation accuracy (N=374, greedy decoding). All fine-grained pruning methods converge near a capability floor ($\sim$75% code capability loss vs Dense), limiting differentiation ($\sim$1 pp range, within stderr). ShortGPT shows a larger GRPO delta (+4.01 pp) but at the 95% CI boundary (stderr $\approx$1.55 pp).

Method	SFT (%)	GRPO (%)	Δ (pp)
Dense	–	53.48	–
CKA-only 20%	12.83	14.17	+1.34
Standard 20%	12.03	13.10	+1.07
RASP (α=1.0) 20%	11.76	13.10	+1.34
ShortGPT 20%†	10.43	14.44	+4.01

†Depth pruning; removes entire layers.

MBPP GRPO deltas are uniformly slightly positive across all fine-grained methods (mean +1.25 pp), but none are statistically significant at this sample size (N=374; stderr $\approx$1.7 pp). This directional consistency may reflect a weak task-specific GRPO effect at the capability floor rather than a violation of reward-accuracy decoupling.

H CKA Scoring Stability

Section 3.2 notes that our intra-layer CKA use avoids known cross-layer sensitivities (Davari et al., 2023; Klabunde et al., 2023). Table 8 validates ranking stability across calibration set sizes.

CKA-based structure rankings are highly stable across calibration data volumes: Spearman ρ between rankings from 256, 512, and 1024 calibration samples ranges from 0.964 to 0.979, confirming that the scoring criterion is robust to calibration set size within this range.

Table 8: CKA scoring stability across calibration sizes. Spearman ρ between structure rankings produced by different calibration set sizes. High correlation ($\rho \geq 0.964$) indicates that CKA-based scoring is robust to calibration data volume.

Calibration Sizes	Spearman ρ (min)	Spearman ρ (mean)
256 / 512 / 1024	0.964	0.979

I Gradient Signal Analysis

We compare per-layer gradient magnitudes during early GRPO training (N=200 GSM8K calibration samples) across three pruning criteria using paired Wilcoxon signed-rank tests.

Table 9: Pairwise gradient magnitude ratios during GRPO training. Ratio: mean gradient magnitude of method A divided by method B across 56 parameter groups (28 layers $\times$ 2 projections). Significant layers: count where paired Wilcoxon signed-rank p<0.05.

Comparison	Gradient Ratio	Sig. Layers
CKA-only vs Standard	2.89$\times$	55 / 56
CKA-only vs RASP	2.26$\times$	56 / 56
RASP vs Standard	1.28$\times$	45 / 56

The gradient magnitude hierarchy (CKA-only $\gg$ RASP $>$ Standard) mirrors the recovery capacity hierarchy (GSM8K: 64.39% $>$ 60.88% $>$ 59.84%), with bootstrap 95% CIs for all ratios excluding 1.0. This is consistent with diversity preservation enabling stronger training signals during GRPO, though the evidence is correlational: an alternative explanation (that CKA-only retains more functional structure, producing higher initial loss and naturally larger gradients) cannot be ruled out.

J Seed Reproducibility

We verify that the recovery capacity spectrum (§4.2) is robust to GRPO random seed variation. CKA-only GSM8K accuracy across 3 GRPO seeds: 64.37% (seed 42), 64.22% (seed 123), 64.59% (seed 456), yielding mean 64.39% $\pm$ 0.19 pp. Standard: mean 59.84% $\pm$ 0.38 pp across the same 3 seeds. The 4.55 pp gap between CKA-only and Standard is stable, with non-overlapping ranges across all seeds. See Table 11 for the full breakdown including 15% sparsity results.

Table 10: Sparsity cliff at 30% structured sparsity. All tested pruning criteria collapse, with near-zero accuracy and EOS generation rates below 6%.

Method	N	Acc. (%)	EOS Rate (%)
Standard	200	4	5.5
RASP (α=1.0)	200	0	3.0
CKA-only	500†	1.6	–†

†Terminated early after collapse confirmed; 8/500 correct.

K Sparsity Cliff at 30%

Section 4.4 notes that all methods collapse at 30% sparsity. Table 10 provides the full results.

All tested methods collapse at 30% sparsity: Standard retains only 4% accuracy, CKA-only 1.6%, and RASP at α=1.0 drops to 0%. This capacity threshold is independent of pruning criterion, validating 20% as the practical operating point for DeepSeek-R1-Distill-Qwen-7B and confirming that the recovery capacity spectrum (§4.2) reflects how methods allocate a limited pruning budget rather than absolute capacity differences.

L Multi-Seed GRPO Results

Table 11: Multi-seed GRPO results on GSM8K (N=1,319). CKA-only, Standard, and Random each trained with 3 GRPO seeds (42, 123, 456); pruning calibration and SFT are identical across all runs within each sparsity level.

Sparsity	Method	Mean Acc. (%)	Std (pp)	Seeds
20%	CKA-only	64.39	0.19	3
	Standard	59.84	0.38	3
	RASP	60.88	–	1
	Random	50.72	0.15	3
15%	CKA-only	72.76	0.57	3
	Standard	65.56	0.50	3

At 20% sparsity, the 4.55 pp gap between CKA-only and Standard is robust across seeds, with non-overlapping ranges (CKA-only: 64.22–64.59%, Standard: 59.44–60.20%). Standard shows a wider observed range across seeds (0.76 pp vs 0.37 pp), consistent with borderline-collapse dynamics where small perturbations produce larger outcome variation. Random (3 seeds: 50.57%, 50.72%, 50.87%; σ=0.15 pp, observed range 0.30 pp) converges in a narrow band with SFT accuracy seed-independent at 51.25%. All three seeds satisfy GRPO $\leq$ SFT (Δ=$-$0.68, $-$0.53, $-$0.38 pp), confirming that reward-accuracy decoupling is reproducible across seeds even for stagnating methods. Note: Standard seed 42 was trained from a separately initialized SFT checkpoint (accuracy 59.21% vs. 59.29% for seeds 123/456; $\Delta = 0.08$ pp). At 15% sparsity, the gap widens to 7.20 pp (72.76% vs 65.56%, $t\approx$16.4, $p<$0.001), with non-overlapping ranges (CKA-only: 72.10–73.09%, Standard: 65.13–65.43%).

M Cross-Sparsity Validation

Table 12: Cross-sparsity validation on DeepSeek-R1-Distill-Qwen-7B. GSM8K accuracy (N=1,319) after SFT warmup and GRPO recovery. 15% and 20% GRPO values are mean over 3 seeds (see Tables 2, 11). ‡Single seed. Mean $|\Delta| \leq 1$ pp across all conditions.

Sparsity	Method	SFT (%)	GRPO (%)	Δ (pp)
15%	CKA-only	72.18	**72.76**±**0.57**	+0.58
	Standard	66.49	65.56±0.50	−0.93
20%	CKA-only	63.91	64.39	+0.48
	Standard	59.29	59.84	+0.55
25%‡	CKA-only	60.12	59.59	−0.53
	Standard	50.87	51.86	+0.99
30%	All	Collapse ($\leq$4%; Table 10)		

At 15% sparsity (3-seed mean), the CKA-only advantage widens to +7.20 pp (72.76% vs 65.56%), compared to +4.55 pp at 20%. CKA-only at 15% (72.76%) represents the highest mean accuracy across all experimental conditions, retaining 80.1% of original dense model performance (90.83%). At 25% sparsity (single seed), Standard's SFT accuracy decline accelerates ($-$7.20 pp from 15%→20%, $-$8.42 pp from 20%→25%), while CKA-only's decline decelerates ($-$8.27 pp then $-$3.79 pp). Reward-accuracy decoupling is preserved ($|\Delta| \leq 1$ pp for both methods) despite Standard's near-zero peak reward (0.008).

N Cross-Scale Validation

To test whether the recovery capacity spectrum generalizes beyond the 7B scale (§4.5), we replicate the comparison on DeepSeek-R1-Distill-Qwen-1.5B (DeepSeek-AI, 2025) at 20% MLP sparsity (Table 13).

The three-method ordering is preserved (CKA-only 35.48% $\gg$ RASP 26.16% $>$ Standard 22.29%), consistent with the 7B results (Table 1). Reward-accuracy decoupling also holds at this scale ($|\Delta| < 1$ pp: CKA-only $-$0.15, Standard $-$0.61 pp). The lower absolute accuracy reflects

Table 13: Cross-scale validation at 20% sparsity. 7B: DeepSeek-R1-Distill-Qwen-7B (GSM8K mean over 3 seeds for CKA-only and Standard; see Table 2). 1.5B: DeepSeek-R1-Distill-Qwen-1.5B (single seed). GSM8K accuracy (N=1,319). 7B reward = peak across 100 GRPO steps (cf. Table 1); 1.5B reward = mean across training steps.

Method	7B		1.5B	
	Peak Rew.	GSM8K (%)	Rew.	GSM8K (%)
CKA-only	**0.354**	**64.39**	**0.238**	**35.48**
Standard	0.081	59.84	0.084	22.29
RASP (α=1.0)	0.061	60.88	0.099	26.16
Random	0.190	–	–	–

the 1.5B model's weaker baseline (49.81% dense vs. 90.83% at 7B), but the relative ordering and decoupling pattern are stable.

O Cross-Model Validation within Qwen2.5 Family

Table 14: Cross-model validation on Qwen2.5-Math-7B (Yang et al., 2024a) (same Qwen2.5 architecture family) at 20% structured sparsity. Accuracy evaluated on GSM8K (N=200). Post-SFT reports accuracy ± binomial standard error (N=200). EOS rate: fraction of generations that produce a valid end-of-sequence token.

Method	Post-Prune (%)	Post-SFT (%)	EOS Rate (%)
CKA-only	1.0	**51.50 ± 3.53**	27.0
Standard	18.5	21.00 ± 2.88	36.5

CKA-only pruning causes near-total initial damage on Qwen2.5-Math-7B (1.0% post-prune accuracy vs Standard's 18.5%), yet SFT recovers CKA-only to 51.50% while Standard remains at 21.00%. This post-prune accuracy reversal provides strong evidence that recovery capacity is shaped by preserved network structure rather than starting accuracy. EOS rates remain below 40% for both methods, indicating that SFT warmup is less effective on this architecture; GRPO validation is deferred to future work.[2]

P Structural Overlap Analysis

CKA-only and Standard pruning produce markedly different removal patterns across the 28 layers of DeepSeek-R1-Distill-Qwen-7B at 20% sparsity. CKA-only concentrates pruning in early layers (L0–9 account for 72.5% of removed MLP groups), preserving nearly all structures in later layers (L16–27). Standard distributes removals more uniformly, pruning substantial numbers of groups in late layers (L16–18, L22, L25–26).

The methods share only partial overlap in which structures they remove. 65 MLP groups are pruned exclusively by CKA-only (retained by Standard), concentrated in L6–9 and L27; these are structures that gradient importance scores as important but CKA identifies as redundant with neighboring groups. Conversely, 46+ groups are pruned exclusively by Standard (retained by CKA-only), concentrated in L16–18, L22, and L25–26; these are structures that gradient importance scores as unimportant but CKA identifies as representationally unique.

This pattern is consistent with the layer-wise CKA similarity analysis (Figure 7): CKA-only's aggressive early-layer pruning disrupts dense representations (low diagonal CKA, μ=0.62) but preserves diverse late-layer pathways, while Standard's uniform pruning maintains dense fidelity (μ=0.90) at the cost of removing unique computational pathways in later layers.

[2]Preliminary GRPO evaluation yields 48.60% (243/500), but 83.8% of responses were truncated at 2048 tokens, precluding direct comparison with SFT results.

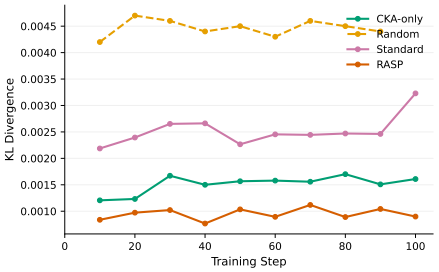


Figure 9: KL divergence between the policy and reference model during GRPO training. RASP maintains near-zero KL (<0.001), consistent with minimal policy movement during collapse. Random shows the highest KL (∼0.004–0.005), indicating more policy exploration. CKA-only and Standard fall between these extremes.

Q KL Divergence During GRPO Training

KL divergence between the evolving policy and the frozen reference model provides a complementary view of the training dynamics reported in Section 4.2. Figure 9 tracks this divergence across GRPO training for all four pruning criteria.

RASP at α=1.0 maintains near-zero KL (<0.001) throughout training, consistent with the reward collapse and vanishing gradients observed in Figure 5: the policy barely moves from its initialization. CKA-only and Standard show moderate

KL (0.001–0.003), while Random exhibits the highest divergence (∼0.004–0.005) despite stagnating reward, suggesting undirected exploration without learning. The KL ordering does not match the recovery capacity spectrum, confirming that policy movement alone does not predict recovery quality.

R Depth-First Pruning Baseline

CKA-only concentrates pruning in early layers (Section 4.5; Appendix P), raising the question of whether its advantage reflects mere depth bias rather than diversity preservation. As a sanity check, we construct a depth-first baseline: synthetic importance scores force removal of late layers first (L18–27: 0.001, L10–17: 0.5, L0–9: 1.0), then standard pruning at 20% sparsity. Both SFT (∼10%) and GRPO (∼8%) accuracy are near chance (early-stopped at $N{\approx}50$–88 samples due to conclusive results), confirming that naive depth ordering cannot substitute for criterion-based structure selection. This baseline operates at layer-level granularity (removing entire transformer blocks), unlike CKA-only/Standard/RASP which prune at MLP channel group level; the granularity difference limits direct comparison with our main results and precludes using this baseline as evidence for the recovery capacity spectrum.

S Per-Layer Capacity Floor Ablation

Table 15: Effect of per-layer capacity floor τ on CKA-only pruning at 20% sparsity. GSM8K accuracy (N=1,319) after GRPO recovery (100 steps). Default τ=0.4.

Floor τ	GRPO Acc. (%)	Δ vs τ=0.4 (pp)
0.3	65.05	+0.68
0.4 (default)	64.37	—
0.5	61.71	−2.66
0.6	62.02	−2.35

The per-layer capacity floor τ sets the minimum fraction of MLP channel groups retained in each layer, preventing over-pruning of any single layer regardless of its importance score. We ablate $\tau \in \{0.3, 0.4, 0.5, 0.6\}$ using CKA-only pruning at 20% sparsity. Lowering τ to 0.3 slightly improves accuracy (+0.68 pp), while raising it to 0.5 or 0.6 causes a notable drop (−2.66 and −2.35 pp respectively; the difference between 0.5 and 0.6 is not significant at 0.31 pp $\ll$ stderr $\approx$ 1.3 pp). Over-constraining per-layer retention hurts global pruning quality by forcing removal of important structures in unconstrained layers.

T TOST Equivalence Testing

To formally assess whether SFT and GRPO produce equivalent GSM8K accuracy, we apply Two One-Sided Tests (TOST) on N=1,319 paired per-example outcomes for CKA-only and Standard at 20% sparsity across multiple GRPO seeds. For each pair, $d_i = \mathbb{1}[\text{SFT correct}]_i - \mathbb{1}[\text{GRPO correct}]_i$, and we test H_0: $|\bar{d}| \geq \varepsilon$ against H_A: $|\bar{d}| < \varepsilon$.

Table 16: TOST equivalence results (ε=2 pp). All observed $|\Delta|$<1 pp. †Separate SFT checkpoint (see Appendix L). ‡300-step extended training (Appendix E).

Criterion	GRPO variant	Δ (pp)	p_{TOST}
CKA-only	seed 42	−0.45	0.008
CKA-only	seed 123	−0.30	0.005
CKA-only	seed 789	−0.15	0.001
CKA-only	seed 456	−0.68	0.023
Standard	seed 123	−0.68	0.020
Standard	seed 456	−0.99	0.069
Standard	seed 42†	−0.23	0.010
Standard	300-step‡	−0.08	0.007

We select ε=2 pp because it is approximately 4× the observed cross-seed standard deviation (σ=0.19 pp for CKA-only at 20% sparsity), ensuring the margin reflects meaningful practical differences rather than measurement noise. At ε=2 pp, 7/8 comparisons reject the null (p<0.05); the remaining pair (Standard vs. seed 456, Δ=−0.99 pp) is borderline (p=0.069). At ε=3 pp, all 8/8 are significant (p<0.002). At ε=1 pp, statistical power is insufficient with binary outcomes at N=1,319.

U Discussion

GRPO as diagnostic probe. The decoupling pattern suggests that GRPO may serve as a *diagnostic probe* for pruning quality at moderate sparsity: reward trajectories appear to indicate whether the pruned architecture retains sufficient diversity for RL exploration. This information is not accessible through SFT, which converges to similar accuracy at 20–25% MLP sparsity. At 15% sparsity, GRPO amplifies the CKA-guided advantage (+1.51 pp), suggesting the transition boundary warrants future investigation.

Implications. For deployment pipelines that include post-pruning RL, diversity-preserving crite-

ria (CKA-only or high-α RASP) should be preferred. Our findings complement Ji et al. (2025): while they demonstrate RLVR superiority for unstructured magnitude pruning at 96.5% retention, within our experimental scope (20–25% structured MLP sparsity on Qwen2.5-family models), this hierarchy dissolves regardless of pruning criterion. The discrepancy likely reflects differences in experimental setup (pruning granularity, sparsity level) rather than a fundamental property of RL recovery.

V Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform the related-work section.